

Disproof of conjecture 00000008338:
the optimal Siegel constant for 1×2 systems is H , not
 $(nH)^{m/(n-m)} = 2H$

Jackmeson1

2026-10-05

1 The conjecture and the clauses refuted

Statement (English, verbatim). Definition: The elimination constant of Siegel’s lemma for linear forms: the height upper bound `sieg_c` of minimal solutions of integer-coefficient linear systems. Conjecture: `sieg_c` (the optimal constant of Siegel’s lemma) is the Bombieri-Vaaler explicit $(nH)^{m/(n-m)}$ (a BV explicit law); tightness of the constant (attainment of minimal solution heights) holds for random integer matrices (a random matrix tightness law); the distribution of shortest kernel vectors of random matrices (Edelman’s kernel distribution) has exponential tails (an exponential tail law); and the tail constant is the half-width $1/(2H)$ (a half-width tail law). (The Chinese text states the same: `sieg_c` is the upper bound of the heights of minimal solutions, and `sieg_c`, the optimal constant of Siegel’s lemma, equals $(n \cdot H)^{m/(n-m)}$.)

Reading. The Definition line names `sieg_c` as *the height upper bound of minimal solutions*: for $m \times n$ integer systems $Ax = 0$ ($0 < m < n$) with coefficients bounded by H , it is the supremum, over all such systems, of the least height of a nonzero integer solution. The parenthetical “the optimal constant” agrees. The conjecture is a conjunction. We refute

(C1) $\text{sieg}_c(m, n, H) = (nH)^{m/(n-m)}$ for all $0 < m < n$ and $H \geq 1$;

(C2) the attainment clause: some (random) integer system has a minimal solution of height $(nH)^{m/(n-m)}$.

Both fail at $(m, n) = (1, 2)$ for *every* $H \geq 1$. Since the ratio is $H/(2H) = 1/2$ for all H , C1 also fails under the reading “asymptotically equal as $H \rightarrow \infty$ ”. We do not address the two tail clauses (Edelman’s kernel distribution and the constant $1/(2H)$); they are not needed, because the conjunction falls with C1.

Readings that are not refuted. (a) If the first clause only meant that $(nH)^{m/(n-m)}$ is a *valid* upper bound, it is Siegel’s lemma and is true. (b) If “tightness” only meant that the *exponent* $m/(n-m)$ cannot be lowered (with an unspecified constant in front), our example does not refute it: at $(1, 2)$ the exponent is 1 and $\text{sieg}_c(1, 2, H) = H$, so the exponent is sharp there. Only the constant factor n is wrong.

Remark on the name. According to the retrieved Wikipedia article on Siegel’s lemma (https://en.wikipedia.org/wiki/Siegel%27s_lemma), for M equations in $N > M$ unknowns “where the coefficients are integers, not all 0, and bounded by B . The system then has a solution” “with the X s all integers, not all 0, and bounded by” $(NB)^{M/(N-M)}$; the same article (https://en.wikipedia.org/wiki/Siegel%27s_lemma) continues “Bombieri & Vaaler (1983) gave the

following sharper bound for the X's:" (a bound in terms of $\det(AA^T)$ and the gcd of the maximal minors). So the formula $(nH)^{m/(n-m)}$ is Siegel's bound, not the Bombieri–Vaaler bound. We test the formula as written in the conjecture.

2 Definitions (as formalized)

All heights are sup norms, and $0 \in \mathbb{N}$.

Definition 1. For $x \in \mathbb{Z}^n$, $\text{ht}(x) = \max_i |x_i|$ (`vecHeight`). For an integer matrix $A \in \mathbb{Z}^{m \times n}$, $\text{ht}(A) = \max_{i,j} |a_{ij}|$ (`matHeight`).

Definition 2. A is *admissible* for H (`Admissible`) if $A \neq 0$ and $\text{ht}(A) \leq H$ (coefficients integers, not all 0, bounded by H , as in the statement of Siegel's lemma quoted above).

Definition 3. $\text{kerHeights}(A) = \{\text{ht}(x) : x \in \mathbb{Z}^n, x \neq 0, Ax = 0\}$ and the height of a minimal solution is $\text{minSolHeight}(A) = \inf \text{kerHeights}(A)$ (Lean `sInf` on $\mathbb{N}$, whose value on $\emptyset$ is 0; we only use it where we prove the set nonempty).

Definition 4. $\text{siegC}(m, n, H) = \sup\{\text{minSolHeight}(A) : A \in \mathbb{Z}^{m \times n} \text{ admissible for } H\}$ (Lean `sSup` on $\mathbb{N}$; for $(1, 2)$ we prove the set has a greatest element, so no junk value is involved). The claimed value is $\text{bvValue}(m, n, H) = (nH)^{m/(n-m)}$, a real power (`Real.rpow`).

3 Proof

Lemma 5 (`exists_kerHeight_le, minSolHeight_le`). *Let $H \geq 1$ and $A = \begin{pmatrix} a & b \end{pmatrix} \in \mathbb{Z}^{1 \times 2}$ with $\text{ht}(A) \leq H$. Then $Ax = 0$ has a nonzero integer solution of height at most H , so $\text{minSolHeight}(A) \leq H$.*

Proof. If $a = b = 0$, take $x = (1, 0)$, of height $1 \leq H$. Otherwise $x = (b, -a) \neq 0$ satisfies $ab + b(-a) = 0$ and $\text{ht}(x) = \max(|a|, |b|) \leq H$. $\square$

This lemma does not use $A \neq 0$, so dropping the condition “not all 0” does not change any of the values below.

Lemma 6 (`le_vecHeight_of_witness, minSolHeight_witness`). *Let $H \geq 1$ and $W = \begin{pmatrix} H & H-1 \end{pmatrix}$ (*witnessMatrix*). W is admissible for H , and $\text{minSolHeight}(W) = H$.*

Proof. $W \neq 0$ and $\text{ht}(W) = H$. If $Hx_0 + (H-1)x_1 = 0$ then $x_1 = H(x_0 + x_1)$, so $H \mid x_1$. If $x_1 = 0$ then $Hx_0 = 0$, so $x_0 = 0$. Hence every nonzero solution has $x_1 \neq 0$, so $\text{ht}(x) \geq |x_1| \geq H$. Combined with the previous lemma, $\text{minSolHeight}(W) = H$. $\square$

Theorem 7 (`isGreatest_siegValues, siegC_one_two, isLeast_real_bound`). *For every $H \geq 1$, H is the greatest minimal-solution height among admissible 1×2 systems, so $\text{sieg}_c(1, 2, H) = H$. Moreover H is the least real B with $\text{minSolHeight}(A) \leq B$ for all admissible A .*

Lemma 8 (`bvValue_one_two`). $\text{bvValue}(1, 2, H) = (2H)^{1/(2-1)} = 2H$.

Theorem 9 (`conjecture8338_false`). *For every $H \geq 1$: $\text{sieg}_c(1, 2, H) = H$, $\text{sieg}_c(1, 2, H) < (2H)^{1/(2-1)}$, $\text{sieg}_c(1, 2, H) \neq (2H)^{1/(2-1)}$, and for every admissible 1×2 system A , $\text{minSolHeight}(A) \neq (2H)^{1/(2-1)}$ (*minSolHeight_lt_bvValue* gives $<$).*

Proof. By the theorem above, $\text{sieg}_c(1, 2, H) = H < 2H$ since $H \geq 1$. Every admissible A has $\text{minSolHeight}(A) \leq H < 2H$. $\square$

The last part refutes C2 for every distribution on 1×2 integer matrices (random or not): no admissible system attains the claimed value. The corollary `not_siegC_eq_bvValue` states that $\forall m, n, H$ with $0 < m < n, H \geq 1$: $\text{sieg}_c(m, n, H) = (nH)^{m/(n-m)}$ is false.

4 Lean formalization and axiom audit

File `lean/Conjecture8338/Basic.lean` (243 lines, `import Mathlib only, namespace Conjecture8338`), Lean v4.33.1, Mathlib v4.33.1. Main statement:

```
theorem conjecture8338_false (H : N) (hH : 1 <= H) :
  siegC 1 2 H = H /\ (siegC 1 2 H : R) < bvValue 1 2 H /\
  (siegC 1 2 H : R) != bvValue 1 2 H /\
  forall A : Matrix (Fin 1) (Fin 2) Z, Admissible H A ->
    (minSolHeight A : R) != bvValue 1 2 H
```

(ASCII transliteration: N, Z, R are $\mathbb{N}, \mathbb{Z}, \mathbb{R}$.) `#print` axioms for `conjecture8338_false`, `not_siegC_eq_bvValue` and `isLeast_real_bound` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`.